KARTHIK CHAKALA

Computer Science Undergraduate | Software Engineering Intern Candidate
B Kothakota, Andhra Pradesh | +91 8328134131 | karthikc11105@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

Full Stack Developer and CS undergraduate with hands-on experience leading end-to-end development of scalable, secure web systems. Proven ability in backend architecture, API design, and real-time systems, complemented by strong algorithmic problem-solving. Seeking SDE Intern roles focused on high-impact, production-grade software.

SKILLS

- Programming Languages (C++, Python, JavaScript)
- Backend & Systems (Node.js, Express, REST APIs, Auth, Docker)
- Databases (PostgreSQL/Supabase, MongoDB, SQLite, MySQL)
- Frontend (React, Next.js, Redux, Tailwind)
- Tools (Git, Docker, Postman)
- Data Structures and Algorithms (Solved 300+ problems on LeetCode and GeeksforGeeks)

PROJECT EXPERIENCE

Hospital Management System (HMS) | Project Lead
Indian Institute of Information Technology, Kurnool | [JUL 2025 – OCT 2025]

- Led a 4-member engineering team to architect and deploy a full-stack Hospital Management System following Agile development practices.
- Built the backend using Node.js, Express, TypeScript, Supabase, and developed the frontend with Next.js and React.
- Engineered 30+ role-based REST APIs supporting appointments, EMR, billing, lab/pharmacy operations, inventory, analytics, and home-visit workflows.
- Integrated Cloudinary, Multer, Nodemailer, and Razorpay, securing 20+ critical endpoints using JWT-based RBAC.

Full-Stack Learning Platform

- Designed and built a full-stack learning platform supporting 5+ core modules including authentication, payments, lessons, quizzes, and competitions.
- Architected a Node.js + Express backend with 40+ REST APIs supporting role-based access for 3+ user roles.
- Implemented real-time competition handling using Socket.IO namespaces and transactional course purchase workflows.
- Enforced security and data integrity using JWT RBAC, HMAC Razorpay verification, parameterized SQL queries, and 10+ relational constraints and triggers.

Flight Fare Prediction System (Machine Learning Research Project)

- Built a flight fare prediction system using a dataset with 300K+ records and evaluated 10 machine learning models.
- Achieved a best R^2 score of 98.47% using Extra Trees Regressor, outperforming prior IEEE benchmarks by ~19%.
- Engineered 15+ predictive features including the Days-Left temporal feature.
- Applied preprocessing, scaling, and K-fold cross-validation to improve model generalization.

EDUCATION

- Indian Institute of Information Technology, Kurnool
  B.Tech in Computer Science and Engineering (2023–2027) | CGPA: 8.4 / 10
- Amaravathi Jr college | MPC | JEE Percentile: 97.4%

RELEVANT COURSEWORK

• Data Structures and Algorithms • Operating Systems • Database Management Systems • Computer Networks • Object-Oriented Programming

ALGORITHMIC PROFICIENCY & CERTIFICATIONS

- Solved 300+ DSA problems across LeetCode and GeeksforGeeks
- LeetCode Rating: 1600+
- CodeChef: 3⋆ | Participated in 20+ contests
- IITB (Python,C,Linux), NPTEL(cloud Computing)